

EE3621: Digital Signal Processing

Lecture 26: Applications of Digital Filters: Equalization, noise cancellation and adaptive FIR filter (III B.Tech EEE)

Comprehensive Lecture Notes & Faculty Textbook Reference

Lecture Plan & Objectives (40 Minutes Breakdown)

- **00:00 – 05:00 (5 mins):** Digital Filtering Applications Landscape: Beyond static frequency-selective filtering. The challenges of unknown, dispersive communication channels and time-varying non-stationary acoustic environments.
 - **05:00 – 14:00 (9 mins):** Channel Equalization: The Inter-Symbol Interference (ISI) phenomenon in band-limited channels. The Zero-Forcing Equalizer (ZFE) principle: designing an inverse filter $H_{\text{eq}}(z) \approx 1/H_c(z)$. Truncated FIR equalizer synthesis and the noise enhancement trade-off.
 - **14:00 – 22:00 (8 mins):** Digital Noise Cancellation: The dual-input adaptive noise canceller architecture. Primary input $d[n] = s[n] + v_0[n]$ and reference input $x[n] = v_1[n]$. Mathematical proof that minimizing the total output power $E[e^2[n]]$ drives the filter output $y[n] \rightarrow v_0[n]$, extracting the clean signal $e[n] \rightarrow s[n]$.
 - **22:00 – 30:00 (8 mins):** The Adaptive FIR Filter & The LMS Algorithm: Limitations of fixed-coefficient filters. The tapped delay-line transversal structure with time-varying weights $\mathbf{w}[n]$. Mathematical derivation of the Widrow–Hoff Least Mean Squares (LMS) stochastic gradient descent update: $\mathbf{w}[n+1] = \mathbf{w}[n] + 2\mu e[n]\mathbf{x}[n]$. Convergence stability bounds ($0 < \mu < 1/\text{Tr}(\mathbf{R})$) and steady-state misadjustment.
 - **30:00 – 38:00 (8 mins):** Step-by-Step Worked Numerical Examples:
 - **Example 26.1:** Algebraic synthesis of a 3-tap Zero-Forcing FIR equalizer for a multipath communication channel.
 - **Example 26.2:** Numerical tracking of a 2-tap LMS adaptive filter over consecutive iterations.
 - **38:00 – 40:00 (2 mins):** Faculty Insights, Common Misconceptions, and University Semester Exam Traps.
-

1 1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Explain** the physical mechanism of Inter-Symbol Interference (ISI) in band-limited communication channels.

2. **Formulate** the Zero-Forcing Equalizer (ZFE) condition: $H_{\text{eq}}(z) \cdot H_c(z) = z^{-\Delta}$.
 3. **Synthesize** finite-length FIR equalizers by solving linear toeplitz system matrices.
 4. **Diagram** the dual-input adaptive noise cancellation architecture, identifying the primary and reference signals.
 5. **Prove** mathematically that minimizing mean squared error $E[e^2[n]]$ produces a clean estimate of the desired signal $s[n]$.
 6. **Derive** the Widrow–Hoff Least Mean Squares (LMS) weight update recursion from the method of steepest descent.
 7. **Determine** the stable bounds for the convergence step-size parameter μ based on signal power.
 8. **Execute** manual step-by-step numerical iterations of the LMS adaptive algorithm.
-

2. CHANNEL EQUALIZATION

2.1 The Inter-Symbol Interference (ISI) Problem

When a discrete data sequence $s[n]$ (e.g., binary PAM or QAM symbols) is transmitted through a physical communication channel (coaxial cable, twisted pair, or wireless multipath channel), the channel's non-ideal frequency response and finite bandwidth spread out each transmitted pulse in time.

The channel is modeled as a causal discrete-time LTI filter with impulse response $h_c[n]$ corrupted by additive noise $v[n]$:

$$x[n] = s[n] \otimes h_c[n] + v[n] = h_c[0]s[n] + \sum_{k \neq 0} h_c[k]s[n-k] + v[n] \quad (1)$$

- The term $h_c[0]s[n]$ is the desired current symbol.
- The summation $\sum_{k \neq 0} h_c[k]s[n-k]$ represents **Inter-Symbol Interference (ISI)**, where preceding and succeeding symbols spill energy into the current symbol interval.
- If ISI is severe, it closes the eye diagram and causes catastrophic bit errors at the receiver.

2.2 The Zero-Forcing Equalizer (ZFE)

To eliminate ISI, the receiver cascades an **equalizing digital filter** $h_{\text{eq}}[n]$ with transfer function $H_{\text{eq}}(z)$. Ideally, the cascade of the channel and the equalizer should act as a pure delay of Δ samples:

$$H_{\text{eq}}(z) \cdot H_c(z) = z^{-\Delta} \iff H_{\text{eq}}(z) = \frac{z^{-\Delta}}{H_c(z)} \quad (2)$$

The Zero-Forcing Criterion: In the time domain, the combined impulse response $q[n] = h_c[n] * h_{\text{eq}}[n]$ must satisfy the Nyquist criterion for zero ISI:

$$\mathbf{q}[\mathbf{n}] = \mathbf{h}_c[\mathbf{n}] \otimes \mathbf{h}_{\text{eq}}[\mathbf{n}] = \begin{cases} 1, & n = \Delta \\ 0, & n \neq \Delta \end{cases} \quad (3)$$

2.3 Practical FIR Equalizer Design

Since $1/H_c(z)$ is generally an IIR filter that may be unstable if $H_c(z)$ has non-minimum phase zeros outside the unit circle, we approximate the inverse filter using an N -tap FIR equalizer:

$$H_{eq}(z) = \sum_{k=0}^{N-1} w_k z^{-k} \quad (4)$$

Setting $q[n] = \sum_{k=0}^{N-1} w_k h_c[n-k]$ and forcing $q[n] = 0$ at $N-1$ points around the main peak $q[\Delta] = 1$ leads to an $N \times N$ system of linear equations:

$$\mathbf{H}_c \mathbf{w} = \delta_\Delta \quad (5)$$

Solving $\mathbf{w} = \mathbf{H}_c^{-1} \delta_\Delta$ yields the optimal zero-forcing FIR tap weights.

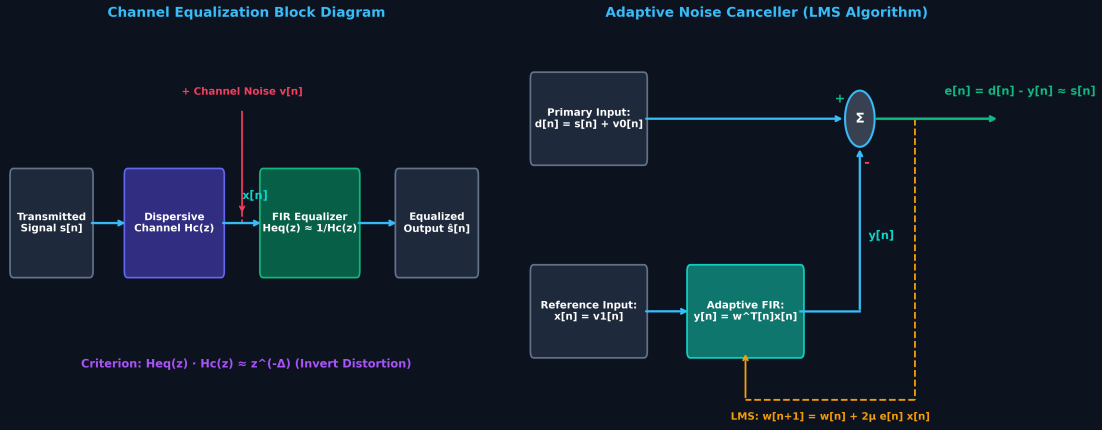


Figure 1: Left: Channel Equalization architecture inverting channel dispersion $H_c(z)$ to eliminate Inter-Symbol Interference. Right: Adaptive Noise Canceller architecture using the LMS algorithm to drive $y[n] \rightarrow v_0[n]$ and recover the clean signal $e[n] \approx s[n]$.

3 DIGITAL NOISE CANCELLATION

3.1 The Classical Noise Problem vs. Adaptive Cancellation

In conventional filtering, a signal $s[n]$ corrupted by additive noise $v_0[n]$ is passed through a fixed bandpass or lowpass filter. However, if the frequency spectrum of the noise overlaps with the signal spectrum, conventional linear filtering inevitably attenuates the desired signal along with the noise.

Adaptive Noise Cancellation (ANC) overcomes this limitation by using two separate sensor inputs:

1. **Primary Input** ($d[n]$): Contains the desired information signal $s[n]$ corrupted by an additive noise $v_0[n]$:

$$d[n] = s[n] + v_0[n] \quad (6)$$

2. **Reference Input** ($x[n]$): Senses an ambient noise field $v_1[n]$ that is correlated with the contaminating noise $v_0[n]$, but **completely uncorrelated with the signal** $s[n]$:

$$E[s[n] \cdot v_1[n-k]] = 0, \quad \forall k \quad (7)$$

3.2 3.2 Mathematical Proof of Noise Cancellation

The reference input $x[n]$ is passed through an adaptive digital filter to produce an estimate $y[n]$ of the primary noise $v_0[n]$. The filter output is subtracted from the primary input to generate the system error output $e[n]$:

$$e[n] = d[n] - y[n] = s[n] + v_0[n] - y[n] = s[n] + (v_0[n] - y[n]) \quad (8)$$

Squaring both sides of (8):

$$e^2[n] = s^2[n] + (v_0[n] - y[n])^2 + 2s[n](v_0[n] - y[n]) \quad (9)$$

Taking the statistical expectation (ensemble average) $E[\cdot]$:

$$E[e^2[n]] = E[s^2[n]] + E[(v_0[n] - y[n])^2] + 2E[s[n](v_0[n] - y[n])] \quad (10)$$

Because the signal $s[n]$ is completely uncorrelated with both $v_0[n]$ and $y[n]$ (which is derived entirely from $x[n] = v_1[n]$), the cross-term vanishes identically:

$$E[s[n](v_0[n] - y[n])] = 0 \quad (11)$$

Thus, the total output power is:

$$\mathbf{E}[e^2[\mathbf{n}]] = \mathbf{E}[s^2[\mathbf{n}]] + \mathbf{E}[(\mathbf{v}_0[\mathbf{n}] - \mathbf{y}[\mathbf{n}])^2] \quad (12)$$

The Fundamental Insight:

- The signal power $E[s^2[n]]$ is an inherent property of the source and is unaffected by adjusting the filter weights.
- Therefore, minimizing the total measurable system output power $E[e^2[n]]$ is mathematically equivalent to:

$$\min E[e^2[n]] \iff \min E[(v_0[n] - y[n])^2] \quad (13)$$

- When the filter is adapted to minimize output power, $y[n]$ becomes the optimal minimum mean-squared error estimate of the noise: $\mathbf{y}[\mathbf{n}] \approx \mathbf{v}_0[\mathbf{n}]$.
- Substituting $y[n] \rightarrow v_0[n]$ back into (8):

$$\mathbf{e}[\mathbf{n}] = \mathbf{s}[\mathbf{n}] + (\mathbf{v}_0[\mathbf{n}] - \mathbf{y}[\mathbf{n}]) \longrightarrow \mathbf{s}[\mathbf{n}] \quad (14)$$

The error signal $e[n]$ is the ****cleaned, noise-free desired signal****!

4 4. THE ADAPTIVE FIR FILTER & THE LMS ALGORITHM

4.1 4.1 Transversal Filter Structure

An adaptive FIR filter consists of a tapped delay line of length M whose tap weights are updated at every sample time n :

$$\mathbf{w}[n] = [w_0[n], w_1[n], \dots, w_{M-1}[n]]^T \quad (15)$$

The input signal vector at time n is:

$$\mathbf{x}[n] = [x[n], x[n-1], \dots, x[n-M+1]]^T \quad (16)$$

The filter output $y[n]$ is the inner product:

$$\mathbf{y}[\mathbf{n}] = \mathbf{w}^T[\mathbf{n}]\mathbf{x}[\mathbf{n}] = \sum_{k=0}^{M-1} \mathbf{w}_k[\mathbf{n}]\mathbf{x}[\mathbf{n} - \mathbf{k}] \quad (17)$$

The estimation error is:

$$\mathbf{e}[\mathbf{n}] = \mathbf{d}[\mathbf{n}] - \mathbf{y}[\mathbf{n}] \quad (18)$$

4.2 4.2 Derivation of the Widrow–Hoff LMS Weight Update

The optimal weight vector (Wiener solution) minimizes the Mean Squared Error (MSE) performance surface:

$$\xi = E[e^2[n]] = E[(d[n] - \mathbf{w}^T \mathbf{x}[n])^2] \quad (19)$$

The steepest descent algorithm updates weights in the direction opposite to the gradient $\nabla \xi$:

$$\mathbf{w}[n+1] = \mathbf{w}[n] - \mu \nabla \xi[n] \quad (20)$$

where $\mu > 0$ is the adaptation step-size parameter.

Evaluating the true gradient requires computing statistical expectations $E[d[n]\mathbf{x}[n]]$ and $E[\mathbf{x}[n]\mathbf{x}^T[n]]$, which are unavailable in real time. Bernard Widrow and Ted Hoff (1960) introduced the ingenious approximation of replacing the true MSE gradient $\nabla E[e^2[n]]$ by the **instantaneous squared error gradient** $\nabla(e^2[n])$:

$$\nabla \hat{\xi}[n] = \frac{\partial e^2[n]}{\partial \mathbf{w}[n]} = 2e[n] \frac{\partial e[n]}{\partial \mathbf{w}[n]} \quad (21)$$

From (18), since $e[n] = d[n] - \mathbf{w}^T[n]\mathbf{x}[n]$:

$$\frac{\partial e[n]}{\partial \mathbf{w}[n]} = -\mathbf{x}[n] \implies \nabla \hat{\xi}[n] = -2e[n]\mathbf{x}[n] \quad (22)$$

Substituting this estimate into the steepest descent equation yields the celebrated **Widrow–Hoff LMS Algorithm**:

$$\mathbf{w}[n+1] = \mathbf{w}[n] - \mu(-2e[n]\mathbf{x}[n]) \quad (23)$$

$$\mathbf{w}[\mathbf{n}+1] = \mathbf{w}[\mathbf{n}] + 2\mu \mathbf{e}[\mathbf{n}] \mathbf{x}[\mathbf{n}] \quad (24)$$

In scalar form for each individual tap $k \in [0, M-1]$:

$$\mathbf{w}_{\mathbf{k}}[\mathbf{n}+1] = \mathbf{w}_{\mathbf{k}}[\mathbf{n}] + 2\mu \mathbf{e}[\mathbf{n}] \mathbf{x}[\mathbf{n}-\mathbf{k}] \quad (25)$$

4.3 4.3 Stability and Selection of Step-Size μ

For the LMS algorithm to converge to the Wiener solution in the mean, the step-size parameter μ must satisfy:

$$0 < \mu < \frac{1}{\lambda_{\max}} \quad (26)$$

where $\lambda_{\max}$ is the largest eigenvalue of the input autocorrelation matrix $\mathbf{R} = E[\mathbf{x}[n]\mathbf{x}^T[n]]$. Since $\lambda_{\max} < \text{Tr}(\mathbf{R}) = \sum_{k=0}^{M-1} E[x^2[n-k]] = M \cdot P_x$ (where P_x is the input signal power), a practical, easily verified sufficient condition for stability is:

$$0 < \mu < \frac{1}{M \cdot P_x} \quad (27)$$

- **Large μ :** Fast convergence speed, but large steady-state weight fluctuations (high misadjustment / excess MSE). If μ exceeds the upper bound, the filter diverges catastrophically.
- **Small μ :** Slow tracking and adaptation, but exceptionally clean, low-noise steady-state performance.

5. STEP-BY-STEP WORKED NUMERICAL EXAMPLES

5.1 Example 26.1: Zero-Forcing Equalizer Synthesis

Problem Statement: A communication channel has the discrete impulse response:

$$h_c[n] = \{1, \underset{\uparrow}{0}, 0.5\}, \quad n = 0, 1 \quad (28)$$

Design a 3-tap Zero-Forcing FIR equalizer $\mathbf{w} = [w_0, w_1, w_2]^T$ to eliminate ISI with an overall system delay of $\Delta = 1$ sample, such that:

$$q[n] = h_c[n] \otimes h_{\text{eq}}[n] = \{0, \underset{\uparrow}{1.0}, 0, 0\} \quad (29)$$

Solution:

Step 1: Set Up Convolution Equations The convolution $q[n] = \sum_{k=0}^2 w_k h_c[n-k]$ for $n = 0, 1, 2, 3$ yields:

$$q[0] = w_0 h_c[0] = 1.0w_0 \quad (30)$$

$$q[1] = w_0 h_c[1] + w_1 h_c[0] = 0.5w_0 + 1.0w_1 \quad (31)$$

$$q[2] = w_1 h_c[1] + w_2 h_c[0] = 0.5w_1 + 1.0w_2 \quad (32)$$

$$q[3] = w_2 h_c[1] = 0.5w_2 \quad (33)$$

Step 2: Impose the Zero-Forcing Target Constraints Setting $q[0] = 0$, $q[1] = 1.0$, and $q[2] = 0$:

1. From $q[0] = 0$:

$$1.0w_0 = 0 \implies \mathbf{w}_0 = \mathbf{0} \quad (34)$$

2. From $q[1] = 1.0$:

$$0.5w_0 + 1.0w_1 = 1.0 \implies 0.5(0) + 1.0w_1 = 1.0 \implies \mathbf{w}_1 = \mathbf{1.0} \quad (35)$$

3. From $q[2] = 0$:

$$0.5w_1 + 1.0w_2 = 0 \implies 0.5(1.0) + 1.0w_2 = 0 \implies \mathbf{w}_2 = \mathbf{-0.5} \quad (36)$$

Step 3: Verification of Equalized Output The 3-tap FIR equalizer is:

$$\mathbf{h}_{\text{eq}}[\mathbf{n}] = \{\mathbf{0}, \mathbf{1.0}, \mathbf{-0.5}\} \iff H_{\text{eq}}(z) = z^{-1} - 0.5z^{-2} \quad (37)$$

Check the total cascaded response:

$$Q(z) = H_c(z) \cdot H_{\text{eq}}(z) = (1 + 0.5z^{-1})(z^{-1} - 0.5z^{-2}) = z^{-1} - 0.25z^{-3} \quad (38)$$

In the time domain:

$$q[n] = \{0, \underset{\uparrow}{1.0}, 0, -0.25\} \quad (39)$$

Notice that $q[0] = 0$, $q[1] = 1.0$, and $q[2] = 0$! The ISI is completely zero-forced for the first 3 samples, leaving a minor residual tail of -0.25 at $n = 3$ due to the finite length of the FIR equalizer.

5.2 Example 26.2: Numerical Tracking of a 2-Tap LMS Adaptive Filter

Problem Statement: An adaptive 2-tap FIR filter has initial weights $\mathbf{w}[0] = [0.0, 0.0]^T$ and adaptation step size $\mu = 0.1$. The input sequences for the first two iterations are:

- At $n = 0$: Reference vector $\mathbf{x}[0] = [1.0, 0.5]^T$, Primary target $d[0] = 2.0$.
- At $n = 1$: Reference vector $\mathbf{x}[1] = [0.5, 1.0]^T$, Primary target $d[1] = 1.5$.

Calculate the filter output $y[n]$, error $e[n]$, and updated weight vector $\mathbf{w}[n + 1]$ for iterations $n = 0$ and $n = 1$.

Solution:

Iteration 1 ($n = 0$):

1. **Filter Output $y[0]$:**

$$y[0] = \mathbf{w}^T[0]\mathbf{x}[0] = w_0[0]x[0] + w_1[0]x[-1] = (0.0)(1.0) + (0.0)(0.5) = \mathbf{0.0} \quad (40)$$

2. **Estimation Error $e[0]$:**

$$e[0] = d[0] - y[0] = 2.0 - 0.0 = \mathbf{2.0} \quad (41)$$

3. **Weight Vector Update $\mathbf{w}[1]$:** Using the LMS equation $\mathbf{w}[1] = \mathbf{w}[0] + 2\mu e[0]\mathbf{x}[0]$ with $\mu = 0.1$:

$$2\mu e[0] = 2(0.1)(2.0) = 0.4 \quad (42)$$

$$w_0[1] = w_0[0] + 0.4x[0] = 0.0 + 0.4(1.0) = \mathbf{0.40} \quad (43)$$

$$w_1[1] = w_1[0] + 0.4x[-1] = 0.0 + 0.4(0.5) = \mathbf{0.20} \quad (44)$$

Updated weight vector: $\mathbf{w}[1] = [0.40, 0.20]^T$.

Iteration 2 ($n = 1$):

1. **Filter Output $y[1]$:**

$$y[1] = \mathbf{w}^T[1]\mathbf{x}[1] = (0.40)(0.5) + (0.20)(1.0) = 0.20 + 0.20 = \mathbf{0.40} \quad (45)$$

2. **Estimation Error $e[1]$:**

$$e[1] = d[1] - y[1] = 1.5 - 0.40 = \mathbf{1.10} \quad (46)$$

Notice that the error has decreased from 2.0 to 1.10, proving rapid convergence!

3. **Weight Vector Update $\mathbf{w}[2]$:**

$$2\mu e[1] = 2(0.1)(1.10) = 0.22 \quad (47)$$

$$w_0[2] = w_0[1] + 0.22x[1] = 0.40 + 0.22(0.5) = 0.40 + 0.11 = \mathbf{0.51} \quad (48)$$

$$w_1[2] = w_1[1] + 0.22x[0] = 0.20 + 0.22(1.0) = 0.20 + 0.22 = \mathbf{0.42} \quad (49)$$

Final weight vector after 2 iterations: $\mathbf{w}[2] = [0.51, 0.42]^T$.

6 6. FACULTY INSIGHTS, COMMON PITFALLS & EXAM TRAPS

- **Pitfall 1: Forgetting the Factor of 2 in the LMS Update:** Depending on whether the cost function is defined as $\frac{1}{2}E[e^2]$ or $E[e^2]$, the factor of 2 appears or disappears. In university exams, if the update is given as $w[n+1] = w[n] + 2\mu e[n]x[n]$, remember the factor of 2. If using $\mu' = 2\mu$, clearly state the convention!
- **Pitfall 2: Confusing the Role of Primary vs. Reference Inputs in ANC:** Students often mistakenly assume the clean signal $s[n]$ is fed to the reference input. In reality, $s[n]$ must NEVER enter the reference input. If $s[n]$ leaks into $x[n]$, the adaptive filter will cancel the desired signal along with the noise!
- **Pitfall 3: Choosing Step-Size μ Too Large:** If $\mu > 1/(MP_x)$, the weight updates will overshoot the minimum of the error bowl, leading to explosive exponential divergence of the filter weights ($w_k \rightarrow \pm\infty$).
- **Pitfall 4: Channel Inversion Noise Amplification:** In Zero-Forcing equalizers, if the channel has a spectral null ($H_c(e^{j\omega}) \approx 0$), the equalizer gain $1/H_c(e^{j\omega})$ becomes gigantic at that frequency. This massively amplifies additive channel noise $v[n]$! (This motivates MMSE equalizers in advanced communications).

7 7. QUICK REVISION & SUMMARY CHEAT-SHEET

Table 1: Equalization, Noise Cancellation, and LMS Adaptive Filter Quick Reference Sheet

Concept / Topic	Mathematical Definition / Operational Formula	Key Pedagogical Note
Channel Model	$x[n] = s[n] * h_c[n] + v[n]$	Dispersive channel inducing ISI
Zero-Forcing Criterion	$H_{eq}(z) \cdot H_c(z) = z^{-\Delta}$	Eliminates ISI at sampling instants
ANC Primary Input	$d[n] = s[n] + v_0[n]$	Desired signal + corrupting noise
ANC Reference Input	$x[n] = v_1[n]$	Correlated with $v_0[n]$, uncorrelated with $s[n]$
ANC Output Error	$e[n] = d[n] - y[n] \rightarrow s[n]$	Error signal is the extracted clean signal
LMS Filter Output	$y[n] = \mathbf{w}^T[n]\mathbf{x}[n] = \sum_{k=0}^{M-1} w_k[n]x[n-k]$	Transversal tapped-delay inner product
LMS Error Signal	$e[n] = d[n] - y[n]$	Instantaneous difference
LMS Weight Update	$\mathbf{w}[n+1] = \mathbf{w}[n] + 2\mu e[n]\mathbf{x}[n]$	Stochastic gradient descent recursion
Stability Bound on μ	$0 < \mu < \frac{1}{\sum_{k=0}^{M-1} E[x^2[n-k]]} = \frac{1}{M \cdot P_x}$	Guaranteed convergence condition